

Persistent homology of percolation complexes on $\{3, q\}$ tilings

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Abstract

We study the homology of percolation clusters on disk-shaped patches of the regular $\{3, q\}$ triangle tilings for $q \geq 7$. Varying q gives a one-parameter family of hyperbolic tilings, from mildly curved ($q = 7$) to deeply hyperbolic ($q = 20$), and this paper measures and derives loop content as a function of curvature. A single growth-rate parameter $\lambda(q)$, the asymptotic ratio of successive ring sizes, governs every result. Total persistent H_1 of the bond-percolation complex, per vertex, is measured across $q = 7, \dots, 20$ and found to be a nearly linear function of $1/\lambda(q)$. An exact identity reduces total loop persistence to the weight of the lattice’s minimum spanning tree minus a boundary correction. The boundary correction follows from the ring recursion, and the spanning-tree weight is derived by attaching rings one at a time to a contracted interior, with a first-order correction for loops that close through the ring outside. The resulting formula has no free parameters, gives the large-curvature intercept in closed form as $5/4 - 2\pi/(3\sqrt{3})$, and reproduces the measured values to within their uncertainties. Whether the same relation between curvature and homology survives once a percolation cluster is embedded into a vector space and fed to a trained model is a natural next question; a preliminary embedding-and-sparse-autoencoder pipeline is sketched as work in progress.

1 Introduction

Hierarchical data is often modeled as, or embedded via, tree structure. Hyperbolic embeddings such as Poincaré or Lorentz embeddings place hierarchies in hyperbolic space specifically to preserve tree distances [Nickel and Kiela, 2017]. With this context, we study the interaction between two considerations that should be taken into account when modeling real data. First, many naturally occurring branching structures are not trees. Connections can exist between downstream nodes independent of heritage. Second, restrictions on possibility and the results of random chance lead us to consider random subgraphs of an underlying graph. Percolation is the process of constructing these random subgraphs. So we study the topology of percolated subgraphs of graphs that embed in hyperbolic space, with a goal being to measure the cost of using an embedding built on the assumption that data are tree-like. Coordinates representing two paths to the same data point, embedded with a tree-like assumption, will disagree by however much the loop’s holonomy fails to close.

This cost is intuitive but rarely turned into a measurable quantity for a specific graph family. This paper does that, using the regular hyperbolic tilings $\{3, q\}$ (equilateral-triangle tessellations with q triangles meeting at every vertex) as a curvature dial. Varying q from 7 upward interpolates between loop-rich and more tree-like percolation behavior, through a single computable growth-rate parameter $\lambda(q)$ (§3). The flat triangular lattice $q = 6$ is the boundary of this family, with $\lambda(6) = 1$; it is excluded here for reasons given in §3.

Section 4 treats the topology of the percolated complex. Bond percolation defines a filtered simplicial complex, and its total persistent H_1 (summed bar length across the whole percolation

process, per vertex) is measured across eleven values of q . The main result is a derivation of that quantity. An exact identity reduces total H_1 persistence to the weight of the lattice’s minimum spanning tree minus a boundary correction. The boundary correction is a consequence of the ring recursion that defines $\lambda(q)$. The spanning-tree weight is derived by attaching rings one at a time to a contracted interior, which reduces it to a percolation calculation on a cycle with spokes, plus a first-order correction for the rare but systematic loops that close through the ring outside. The formula that results has no adjustable parameters and matches the measurements at every q to within their uncertainties. In particular it explains why the measured values are so close to linear in $1/\lambda(q)$: the expansion parameter of the derivation is $1/\lambda(q)$ itself.

Section 2 sets up persistent homology as the diagnostic used throughout. Section 3 defines the family, its growth rate $\lambda(q)$, and the two combinatorial facts about ring structure the derivation depends on. Section 4 contains the derivation and the measurements. Section 5 discusses one concrete application: percolation-based synthetic data models for mechanistic interpretability research, which rely on the same tree-likeness assumption measured here. Section 6 sketches a natural extension: embedding percolated clusters into a vector space and testing whether the dependence of loop content on curvature survives the embedding and shows up in the features a trained model learns from it.

2 Persistent homology as the diagnostic

Bond percolation defines a natural filtration. Given a graph G , assign each edge $e \in G$ an i.i.d. Uniform $[0, 1]$ occupation threshold $u_e \in [0, 1]$, and build the clique (flag) complex of the graph using those thresholds as filtration values. Declare edge e occupied at parameter p whenever its threshold $u_e \leq p$, and declare 2-simplices occupied wherever a triangle’s three edges are all present. A triangle enters the filtration at the maximum of its three edge thresholds. Reusing the same draw of thresholds across all p makes the occupied sets nested: one draw of $\{u_e\}$ gives a single filtration that encodes the entire coupled percolation process across every value of p . Standard persistent homology software (here, GUDHI) extracts persistent H_0 and H_1 from that one filtration directly, as Betti curves $\beta_0(p)$, $\beta_1(p)$ and persistence diagrams whose bars record when each loop is born and when it gets filled in. Any single draw of $\{u_e\}$ is still one sample path, and the resulting curves fluctuate from draw to draw. The measurements below average over 12 independent draws per lattice, with the reported spread reflecting that realization-to-realization variation. This connects to existing work on homological percolation, which studies the birth and death of giant cycles in percolation processes via the Euler characteristic curve on cubical and other lattices [Bobrowski and Skraba, 2020]; the approach here specializes to the $\{3, q\}$ triangulated case and adds curvature as an explicit parameter.

One technical condition makes this tractable: as long as the graph’s only 3-cliques are genuine faces of the underlying triangulation (no “accidental” triangles among mutually adjacent vertices that don’t bound a real 2-cell), the clique complex reconstructs the correct 2-cell structure automatically. This was verified computationally for every lattice used below (Euler characteristic exactly 1 in each case, confirming a topological disk, and an exact match between the graph’s 3-cliques and the known face list, with zero spurious or missing triangles).

One subtlety matters again in §4.1. A single triangular face’s own 3-cycle is born, in this filtration, only once all three of its edges are present, which is also the threshold at which the 2-simplex fills it in. Isolated triangle loops are therefore born and die simultaneously and contribute zero persistence; whatever loop content survives has to come from cycles spanning more than one face. Figure 1 shows what this looks like on $\{3, 7\}$ at four occupation probabilities: the filled

triangles are shaded, and the highlighted cycles are a basis for H_1 of the flag complex at that p , so a fully occupied triangle on its own is never highlighted.

3 The $\{3, q\}$ tilings and their ring structure

The family of regular triangulations $\{3, q\}$ (q equilateral triangles meeting at every vertex) is built by a ring-growth construction: starting from a single vertex, each successive ring is attached according to the fixed combinatorial rule that q triangles meet at every vertex. Ring sizes satisfy the linear recursion

$$rc[n] = (q - 4) \cdot rc[n - 1] - rc[n - 2], \quad rc[0] = 1, \quad rc[1] = q,$$

which has characteristic equation $\lambda^2 = (q - 4)\lambda - 1$ with dominant root

$$\lambda(q) = \frac{(q - 4) + \sqrt{(q - 4)^2 - 4}}{2}.$$

This is the tiling's asymptotic per-ring growth rate, $rc[n]/rc[n - 1] \rightarrow \lambda(q)$ as $n \rightarrow \infty$. For $q \geq 7$ the discriminant is positive and $\lambda > 1$, giving exponential growth. Table 1 gives illustrative values.

q	geometry	$\lambda(q)$
7	mildly hyperbolic	2.618
8	more hyperbolic	3.732
12	strongly hyperbolic	8.873
20	deep hyperbolic	17.944

Table 1: Illustrative values of the growth rate $\lambda(q)$.

Two consequences of the recursion do all the work in §4.

The outermost ring is a fixed fraction of the disk. The total vertex count $V = 1 + \sum_n rc[n]$ is a geometric series dominated by its last term, so the number b of boundary edges (edges bounding only one face) satisfies $b/V \rightarrow 1 - 1/\lambda(q)$ as the ring depth grows. Table 2 shows this is already accurate to four decimal places at $V \sim 10^3$. A disk of any depth is therefore mostly rim: for $q = 7$, 62% of all vertices lie on the outermost ring, and for $q = 20$, 94%.

q	V	b/V	$1 - 1/\lambda(q)$
7	617	0.6240	0.6180
8	2,281	0.7330	0.7321
9	1,306	0.7925	0.7913
12	6,817	0.8731	0.8730
16	2,497	0.9163	0.9161
20	5,441	0.9373	0.9373

Table 2: Boundary-edge fraction versus its large- N limit, at moderate ring depths.

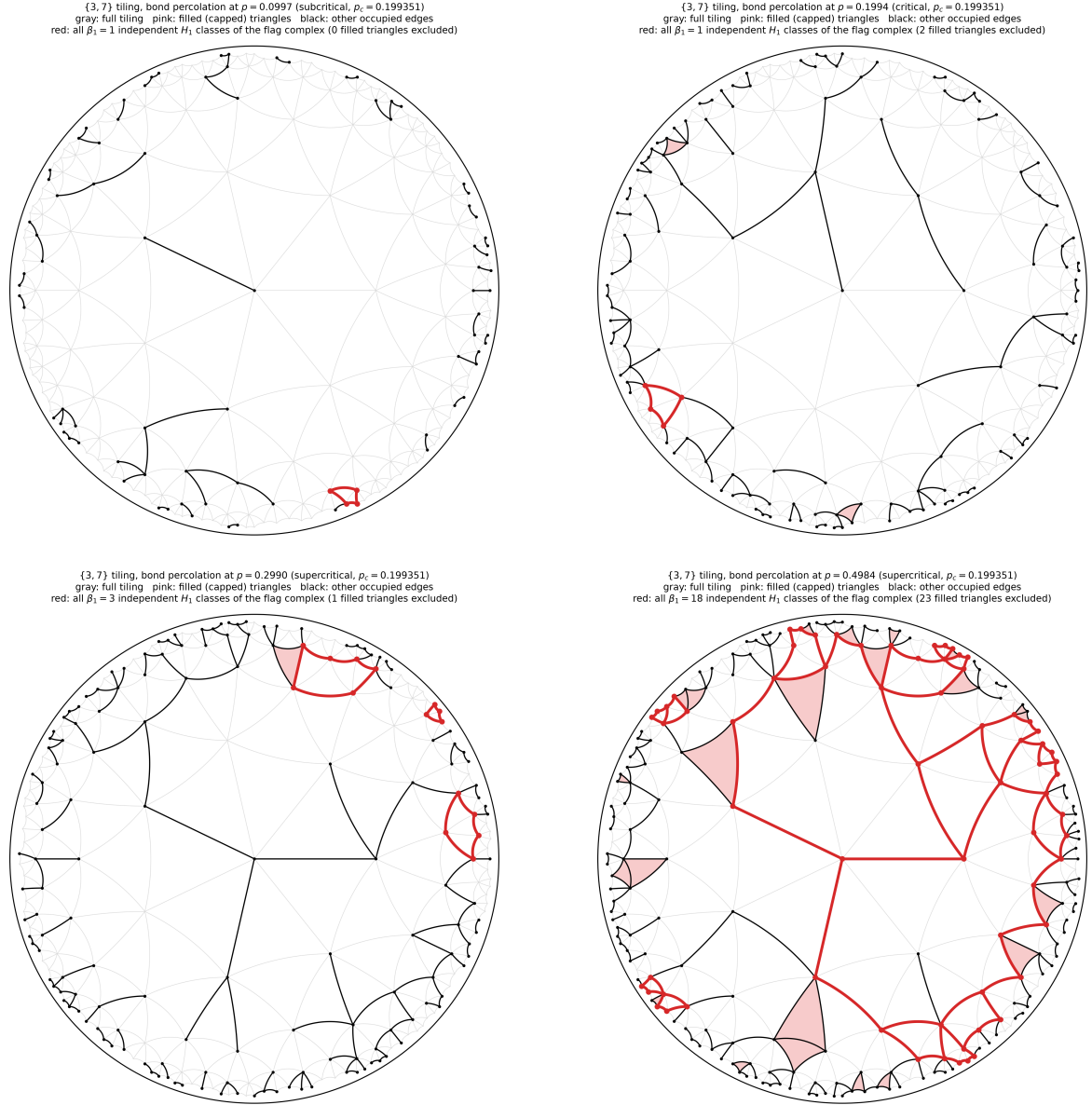


Figure 1: The $\{3, 7\}$ tiling to ring 4 ($V = 232$) in the Poincaré disk, with every edge drawn as a hyperbolic geodesic, under bond percolation at $p = 0.10, 0.20, 0.30$, and 0.50 (top left to bottom right; the bond threshold is $p_c \approx 0.199$). Gray: the full tiling. Black: occupied edges. Pink: triangles with all three edges occupied, which the flag complex fills in. Red: a basis for H_1 of the flag complex, computed as the cycle space of the occupied graph modulo the boundaries of the filled triangles. At this small size the critical draw happens to carry a single loop; loop density per vertex at p_c is small but nonzero, and grows with p well past the threshold, which is the regime that dominates the integrated quantity studied in §4.

Each ring has two kinds of vertex. A vertex in ring n is adjacent either to one vertex of ring $n - 1$ (call it e-type) or to two (v-type). The v-type vertices are the children shared between adjacent parents, and counting them is immediate: ring $n - 1$ is a cycle, so it has $rc[n - 1]$ adjacent pairs, each pair has exactly one common child in ring n , and distinct pairs have distinct common children. Hence

$$v_n = rc[n - 1], \quad \text{so} \quad \frac{v_n}{rc[n]} = \frac{rc[n - 1]}{rc[n]} \longrightarrow \frac{1}{\lambda(q)}. \quad (1)$$

The v-type count is exact at every ring; only the fraction is asymptotic, and it converges at the same rate as the ring ratio itself.

The arrangement of the two types around a ring is rigid rather than random, as the recursive, q -fold symmetric construction requires. The sequence is periodic with period $rc[n]/q$, the fundamental domain of the rotational symmetry, and within a period it is balanced: every window of L consecutive vertices contains either $\lfloor L\alpha_n \rfloor$ or $\lceil L\alpha_n \rceil$ v-types, where $\alpha_n = rc[n - 1]/rc[n]$. (A balanced periodic word of rational slope is a Christoffel word; as n grows, $\alpha_n \rightarrow 1/\lambda(q)$, which is irrational, and these words converge to the Sturmian word of that slope.) Both properties were checked directly on the constructed graphs for every q and ring depth used here.

The derivation in §4.3 treats the type of each vertex as if it were independent with probability α_n , which the balance property says it is not: consecutive v-types never occur, and the gaps between them take only two lengths. Redoing the renewal sum with the balanced counting statistic in place of the binomial one changes the result by less than 10^{-3} at $q = 7$ and by less at larger q , well inside the other approximations, so the independent treatment is kept.

Why $q = 6$ is excluded. At $q = 6$ the recursion has a repeated root $\lambda = 1$: ring sizes grow linearly, the boundary fraction b/V tends to zero rather than to a positive constant, and there is no small parameter. Every result in §4 is an expansion in $1/\lambda(q)$, and at $\lambda = 1$ that expansion has nothing to expand in. The flat case also behaves differently in a purely practical sense: its per-vertex loop content converges to its large- N limit only as a power law in N (roughly $N^{-0.45}$, needing on the order of 10^6 vertices for three-figure accuracy), whereas every hyperbolic q is converged to within 1 to 2% by 10^4 vertices, consistent with the outermost ring being a fixed fraction of the whole so that adding a ring adds the same per-vertex contribution. The flat lattice is the amenable member of the family and is genuinely a different object; it is not part of this paper.

Criticality is not what is being measured. For $q \geq 7$ the tilings are nonamenable and Gromov-hyperbolic, and percolation on them has mean-field critical behavior [Hutchcroft, 2019, Mertens and Moore, 2017]. That is not what governs the quantity studied here. Total persistent H_1 integrates $\beta_1(p)$ over all $p \in [0, 1]$, and almost none of that integral comes from near the percolation threshold: restricting to a window $[p_c/2, 2p_c]$ around the bond threshold captures under 5% of the total at $q = 7$ and under 0.01% by $q = 20$. Loop content in these complexes is a supercritical, bulk phenomenon, and the derivation in §4 never uses critical-point theory. Figure 2 shows a critical draw on a larger disk: loops are present at a small, stable density, and they sit where the vertices are, on the rim.

4 The topological loop content of percolated subsets

4.1 An exact identity

Because the percolation complex is a triangulated topological disk (§2), $\beta_2(p) = 0$ identically: a 2-dimensional complex only carries an H_2 class around a genuine enclosed void, which requires

$\{3, 7\}$ tiling, bond percolation at $p = 0.1994$ (critical, $p_c = 0.199351$)
 gray: full tiling pink: filled (capped) triangles black: other occupied edges
 red: all $\beta_1 = 31$ independent H_1 classes of the flag complex (102 filled triangles excluded)

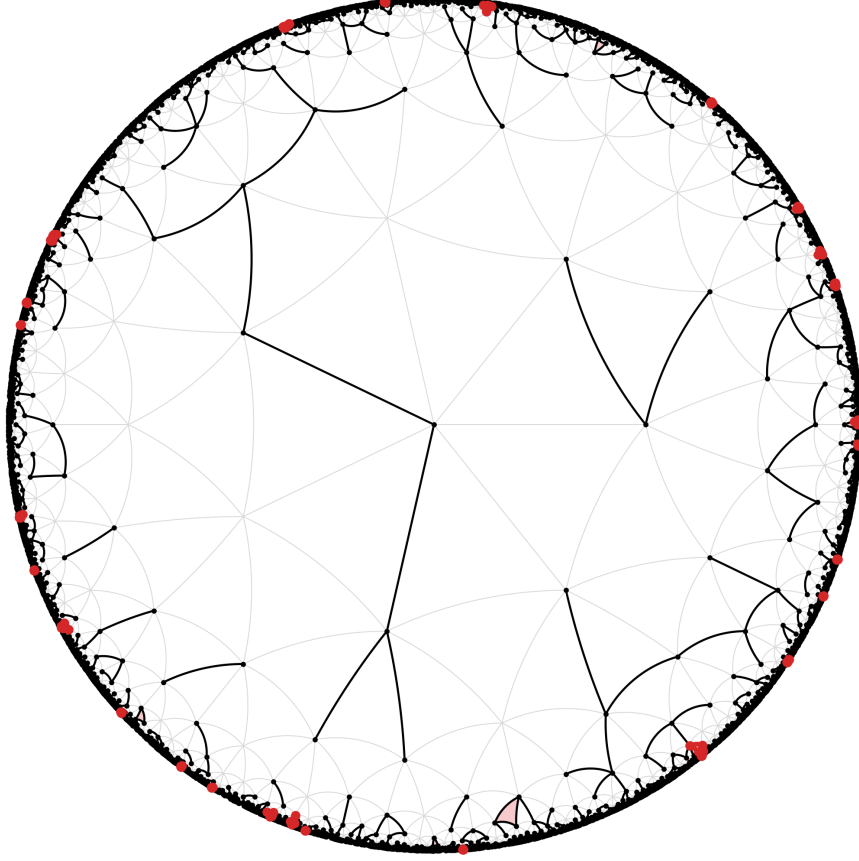


Figure 2: The $\{3, 7\}$ tiling to ring 8 ($V = 11,173$) at $p = p_c$, same color scheme as Figure 1. This draw carries 31 independent H_1 classes; the loop density at p_c is about 0.003 per vertex and is stable from ring 7 onward, so the single loop in the ring-4 panel of Figure 1 is a small-sample effect. The classes sit almost entirely on the outermost ring, which holds 62% of all vertices at $q = 7$. The interior looks empty because it holds few vertices, not because its loop density is lower.

a closed surface, and a disk's boundary is never closed. Consequently the Euler characteristic $\chi(p) = V - E(p) + F(p)$ (vertices are present at every p ; $E(p), F(p)$ count edges and triangles occupied by threshold p) satisfies $\chi(p) = \beta_0(p) - \beta_1(p)$ exactly, at every p , giving

$$\beta_1(p) = \beta_0(p) - V + E(p) - F(p).$$

Total H_1 persistence is $\int_0^1 \beta_1(p) dp$, the area under the loop-density curve. Integrating term by term: writing u_e for edge e 's occupation threshold, $E(p) = \sum_e \mathbb{1}[u_e \leq p]$ gives $\int_0^1 E(p) dp = \sum_e (1 - u_e)$; writing $m_t = \max(\text{the three edge thresholds of triangle } t)$ for its occupation threshold, $\int_0^1 F(p) dp = \sum_t (1 - m_t)$.

The $\beta_0(p)$ term is the classical single-linkage/Kruskal fact: the connected-component structure of the graph restricted to edges with $u_e \leq p$ is, for every p , identical to the component structure using only the edges of the graph's minimum spanning tree (MST, the subset of $V - 1$ edges connecting all vertices at minimum total weight, built greedily by Kruskal's algorithm in increasing weight order) with weight $\leq p$. So $V - \beta_0(p)$ equals the number of MST edges with weight $\leq p$, and $\int_0^1 \beta_0(p) dp = 1 + w_{\text{MST}}$, where $w_{\text{MST}} = \sum_{e \in \text{MST}} u_e$.

Assembling the pieces gives an exact identity, not an approximation and not only in expectation, for a single percolation realization:

$$\int_0^1 \beta_1(p) dp = w_{\text{MST}} - \sum_e u_e + \sum_t m_t. \quad (2)$$

This was checked directly against the GUDHI pipeline: for a $\{3, 7\}$ lattice at ring depth 5 ($V = 617$), the right-hand side of (2) matches GUDHI's computed total H_1 persistence to floating-point precision (10^{-12}).

4.2 Reduction to a spanning-tree weight and a boundary term

Taking expectations, $\mathbb{E}[u_e] = \frac{1}{2}$ and $\mathbb{E}[m_t] = \frac{3}{4}$ (the mean of the maximum of three i.i.d. Uniform $[0, 1]$ variables), so $\mathbb{E}[\sum_e u_e] = E_{\text{tot}}/2$ and $\mathbb{E}[\sum_t m_t] = 3F_{\text{tot}}/4$, where $E_{\text{tot}}, F_{\text{tot}}$ are the lattice's total edge and face counts. Writing b for the number of boundary edges, Euler's formula together with the disk's face structure gives the exact relations $E_{\text{tot}} = 3V - 3 - b$ and $F_{\text{tot}} = 2V - 2 - b$. Substituting into (2) in expectation:

$$\mathbb{E} \left[\int_0^1 \beta_1(p) dp \right] = \mathbb{E}[w_{\text{MST}}] - \frac{b}{4}. \quad (3)$$

With $b/V \rightarrow 1 - 1/\lambda(q)$ from §3, the boundary term contributes exactly $-\frac{1}{4} + \frac{1}{4\lambda(q)}$ to H_1 persistence per vertex. Everything else is in the bulk term

$$\mu(q) := \lim_{V \rightarrow \infty} \frac{\mathbb{E}[w_{\text{MST}}]}{V},$$

the expected weight of a minimum spanning tree per vertex under i.i.d. Uniform $[0, 1]$ edge weights. Because the outermost ring is a fixed fraction of the disk, $\mu(q)$ is a property of the disk sequence, not of a translation-invariant bulk; that is what makes it derivable.

4.3 Deriving the bulk term

The identity $\int_0^1 \beta_0(p) dp = 1 + w_{\text{MST}}$ runs both ways, so $\mathbb{E}[w_{\text{MST}}] = \int_0^1 (\mathbb{E}[C(p)] - 1) dp$ where $C(p)$ is the number of connected components at threshold p . This turns the spanning-tree weight into a percolation question: how many components, on average, at each p .

One ring at a time. Build the disk ring by ring and ask how many new components ring n contributes at threshold p , given everything inside it. If the interior were a single connected component, then ring n 's contribution would be the number of maximal runs of consecutive ring- n vertices, joined by present rim edges, that have no present spoke into the interior. Ring n is a cycle of $rc[n]$ vertices; each vertex has one spoke inward if e-type and two if v-type; and rim edges, spokes, and vertex types are all independent of one another. With $\alpha = 1/\lambda(q)$ the v-type fraction, the probability that a given vertex has no present inward spoke is $(1-p)(1-\alpha p)$, and the runs are a renewal process along the cycle with geometric lengths. Summing over runs gives, per vertex of the ring,

$$c(p; \alpha) = \frac{(1-p)^3(1-\alpha p)}{1-p(1-p)(1-\alpha p)}, \quad \mu_{\text{wheel}}(\alpha) := \int_0^1 c(p; \alpha) dp. \quad (4)$$

Since every ring contributes the same per-vertex amount, $\mu_{\text{wheel}}(\alpha)$ is the bulk density under the single-connected-interior assumption. At $\alpha \rightarrow 0$ the integral is elementary:

$$\mu_{\text{wheel}}(0) = \frac{3}{2} - \frac{2\pi}{3\sqrt{3}} = 0.290800\dots \quad (5)$$

Where the assumption is exact, and where it is not. The outermost ring has nothing outside it, so the only way one of its spoke-free runs can join anything is through the interior, and for it the assumption is exact. An inner ring is different: two of its spoke-free runs, separated by an absent rim edge between vertices u and $u+1$, can be joined through the ring outside, because u and $u+1$ have a common child w there (the v-type vertex they share). If a run of present rim edges through w carries a present spoke to u on one side and to $u+1$ on the other, the two runs are one component, and the single-interior count is one too high. Averaging over the geometric length of the run through w gives the probability that a given absent rim edge is bridged this way,

$$\beta(p) = \left(\frac{p}{1-p+p^2} \right)^2, \quad (6)$$

so that the effective probability two consecutive ring vertices are connected, directly or via the ring outside, is $p_{\text{eff}} = p + (1-p)\beta(p)$. Recomputing the run count with p_{eff} for rim connectivity and p for spoke presence,

$$c_{\text{eff}}(p) = \frac{(1-p_{\text{eff}})^2(1-p)}{1-p_{\text{eff}}(1-p)},$$

and the overcount per inner-ring vertex, integrated over p , is

$$B_1 = \int_0^1 [c(p; 0) - c_{\text{eff}}(p)] dp = 0.04713\dots \quad (7)$$

The integrand is the rational function $p^2(1-p)^3(p^4 - p^3 + p^2 - p + 1)/[(p^2 - p + 1)^2(p^6 - 3p^5 + 5p^4 - 5p^3 + 5p^2 - 3p + 1)]$; the sextic is palindromic and reduces to an irreducible cubic in $p + 1/p$, so B_1 has a closed form, but not an illuminating one. Since the inner rings are a $1/\lambda$ fraction of the disk, and bridging through two or more rings is suppressed by further powers of $1/\lambda$,

$$\mu(q) = \mu_{\text{wheel}}\left(\frac{1}{\lambda(q)}\right) - \frac{B_1}{\lambda(q)} + O\left(\lambda(q)^{-2}\right). \quad (8)$$

Checks on the pieces. Each ingredient was tested separately by simulation on the constructed graphs, not only the final sum. The v-type fraction is $1/\lambda(q)$ to four decimals. The outermost ring’s run count matches $c(p; \alpha)$ to about 1% at every p , and its overcount is exactly zero at every p . The bridging probability $\beta(p)$, measured directly at absent rim edges, matches (6) to within 2% across p . The per-vertex overcount of an inner ring, measured at $q = 20$ and $q = 12$ where the ring outside is long enough for the local picture to apply, integrates to 0.047 and 0.045 against $B_1 = 0.0471$. And the abstract run-counting model reproduces $c - c_{\text{eff}}$ to four digits in its own Monte Carlo. The remaining difference between (8) and measurement scales as λ^{-2} with a coefficient near -0.01 , as a first-order expansion should leave.

4.4 The predicted law and the measurements

Combining (3) and (8), total persistent H_1 per vertex is

$$\frac{H_1}{V} = \mu_{\text{wheel}}\left(\frac{1}{\lambda}\right) - \frac{B_1}{\lambda} - \frac{1}{4} + \frac{1}{4\lambda} + O(\lambda^{-2}), \quad \lim_{\lambda \rightarrow \infty} \frac{H_1}{V} = \frac{5}{4} - \frac{2\pi}{3\sqrt{3}} = 0.04080\dots \quad (9)$$

There are no fitted parameters in (9). The large-curvature intercept is a closed form. The dependence on q enters only through $\lambda(q)$, and the leading behavior is linear in $1/\lambda$ because that is the expansion parameter; μ_{wheel} is not exactly linear in α , which is why a straight line is a good but not perfect description.

Measurements. Total persistent H_1 was measured for $q = 7, 8, 9, 10, 11, 12, 13, 14, 16, 18, 20$, at each q ’s largest tested system size (V between 3,781 and 31,809), averaging 12 percolation draws. The bulk density $\mu(q)$ needs only Kruskal’s algorithm, so it was measured separately at much larger sizes, up to $V \approx 2 \times 10^6$. Tables 3 and 4 compare both to the derived values, and Figure 3 plots them against $1/\lambda(q)$.

q	$\lambda(q)$	measured $\mu(q)$	derived, eq. (8)	difference
7	2.618	0.24557	0.24430	−0.00127
8	3.732	0.25852	0.25796	−0.00056
9	4.791	0.26548	0.26514	−0.00034
10	5.828	0.26990	0.26966	−0.00024
11	6.854	0.27297	0.27279	−0.00018
12	7.873	0.27532	0.27511	−0.00021
13	8.887	0.27700	0.27688	−0.00012
14	9.899	0.27837	0.27830	−0.00007
16	11.916	0.28055	0.28040	−0.00015
18	13.928	0.28198	0.28190	−0.00008
20	15.937	0.28310	0.28302	−0.00008

Table 3: Bulk minimum-spanning-tree weight per vertex, measured at V up to 2×10^6 , against the derived value. Root-mean-square difference 0.00045.

On fitting. Before the derivation existed, the natural summary of Table 4 was a two-parameter fit, $H_1/V \approx 0.0404 + 0.1316/\lambda(q)$ with $R^2 = 0.9995$. A fit that good over eleven points of a smooth monotone function is weaker evidence than it feels: many decreasing functions of q would

q	measured H_1/V	derived, eq. (9)	difference
7	0.0910 ± 0.0009	0.0898	-0.0012
8	0.0755 ± 0.0005	0.0750	-0.0005
9	0.0676 ± 0.0007	0.0673	-0.0003
10	0.0630 ± 0.0010	0.0625	-0.0005
11	0.0594 ± 0.0004	0.0593	-0.0001
12	0.0578 ± 0.0013	0.0569	-0.0009
13	0.0552 ± 0.0004	0.0550	-0.0002
14	0.0535 ± 0.0008	0.0536	+0.0001
16	0.0516 ± 0.0006	0.0514	-0.0002
18	0.0500 ± 0.0015	0.0498	-0.0002
20	0.0488 ± 0.0016	0.0487	-0.0001

Table 4: Total persistent H_1 per vertex, measured (mean and realization-to-realization spread over 12 draws) against the derived value. Root-mean-square difference 0.0005; every point is within its own spread.

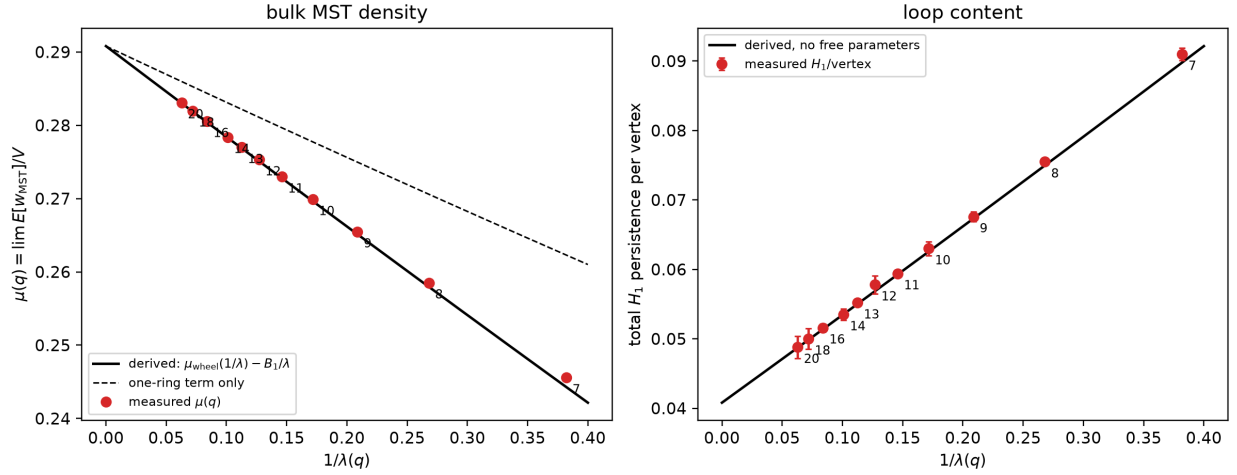


Figure 3: Left: measured $\mu(q)$ against $1/\lambda(q)$, with the derived curve (8) (solid) and the one-ring term μ_{wheel} alone (dashed). Right: measured total H_1 per vertex with the derived curve (9). Neither curve has a fitted parameter.

fit comparably, and $1/\lambda(q)$ could have been an artifact of the range. The derivation removes that concern by producing the functional form rather than assuming it. A straight line fitted to the derived curve (9) over the same q is $0.0406 + 0.1286/\lambda(q)$, which is the empirical fit to within its own uncertainty. The fit is now a check on the derivation, not the result.

4.5 Remarks

The large-curvature limit of the bulk term, $\mu_{\text{wheel}}(0) = 3/2 - 2\pi/(3\sqrt{3})$, should not be read as a Bethe-lattice value. As $q \rightarrow \infty$ the v-type fraction $1/\lambda$ goes to zero, so each ring attaches to the interior through single spokes, but the ring is still a cycle of filled triangles, and the cycle is what the calculation in §4.3 is about. The tiling never becomes the triangle-free regular tree that “Bethe lattice” ordinarily denotes; it is tree-like at long range only, and the nonzero intercept $5/4 - 2\pi/(3\sqrt{3})$ in (9) is the loop content that survives on the rim.

What remains open is the $O(\lambda^{-2})$ term in (8). Numerically it is about $-0.01/\lambda^2$. It collects bridging through two rings and the correlation between bridges at neighboring rim breaks, both of which are local computations of the same kind as (6), and there is no obstacle in principle to carrying the expansion one order further. Whether the full series resums to something closed is a separate question. The same method applies to any $\{p, q\}$ family built by ring growth, with λ , the v-type fraction, and the seam structure replaced by their analogues.

5 Application: percolation-based data models for interpretability research

Percolation-based synthetic data models for mechanistic interpretability research are one concrete setting where the tree-likeness question measured above already matters in practice. Principles of Intelligence’s published data model treats a data distribution as a critical percolation cluster process on a high-dimensional lattice, generated by an explicit hierarchical fine-graining algorithm and embedded into a vector space via a branching random walk, the same embedding construction sketched as future work in §6 [Brill, 2024, 2025]. The high-dimensional case is analytically tractable because, in high dimensions, a random path essentially never self-intersects, so percolation clusters are vanishingly unlikely to contain loops; the lattice can then be replaced by a Bethe lattice, an infinite tree of fixed degree, for which the percolation threshold and critical exponents are solvable exactly [Brill, 2025, Stauffer and Aharony, 1994].

This is a homological claim dressed in probabilistic language: it says H_1 of the percolation cluster vanishes in the relevant limit, leaving only H_0 (connectivity), which is all the tree picture needs. §4 turns “is the treelike approximation good enough” from a qualitative appeal to dimension into a number: total H_1 persistence per vertex is given by (9), a finite, invertible expression rather than an asymptotic statement about $d \rightarrow \infty$. Given a target loop-persistence value, the law inverts, picking off the curvature (or, between integer q , an interpolated tiling family) that hits any specified distance from the treelike limit. The high-dimension argument alone cannot offer that: it only ever says the approximation gets better, not by how much at any particular finite d .

A further check the original data-model literature does not attempt, sketched but not resolved in §6, is whether the same suppression, measured here on the abstract complex, actually reaches a trained model’s learned features once the data is embedded and used for training, rather than remaining a fact about the input distribution’s own combinatorics that a downstream model might or might not pick up on. If it holds, that would be evidence that a data model’s tree-likeness at the input level is something a trained model’s own representations can inherit, not just a modeling

convenience. That check is left to future work here, not claimed as a result.

Two things this paper does not claim. First, it is not a demonstration that Brill’s model is right or wrong in the high-dimensional regime it is actually intended for: for $d \gtrsim 6$, percolation clusters have fractal dimension $4 < d$, and the model’s own asymptotic argument already applies on its own terms, without needing curvature as a substitute. Second, curvature and dimension are not interchangeable in general. What is shown here is that they produce the same qualitative mechanism (tree-likeness driven by an isoperimetric, non-amenability property, rather than by literal high-dimensional sparsity), not that they are numerically equivalent knobs. Curvature’s real advantage is that it is a controllable, finite-size, quantifiable stand-in for a question that is otherwise only answerable in an asymptotic limit: it can be measured, dialed, and now computed on graphs small enough to simulate directly, which the high-dimensional case is not.

6 Further work: embedding percolated subsets into vector space

Section 4 measures H_1 on the abstract percolation complex, a quantity defined without reference to any embedding at all. A natural next question is what happens once a cluster is no longer just an abstract complex but an actual point cloud: does curvature’s suppression of loop content survive being embedded into a vector space and fed to a trained model, the object a downstream model such as a sparse autoencoder (SAE) would actually receive as input?

The natural embedding to try first is a **branching random walk (BRW)**: a breadth-first-search spanning tree of each cluster is chosen, each tree edge is assigned an i.i.d. Gaussian step, and each node’s coordinate is the cumulative sum of steps along its unique path from the cluster’s root. This reproduces the tree’s own path metric exactly, so any embedding-space distortion at non-tree (cycle-closing) edges is directly attributable to the cluster’s own loop content rather than to an artifact of the embedding scheme: a **cycle inconsistency** score, measurable with no homology computation at all. Training a sparse autoencoder on the pooled embeddings and measuring **feature-splitting**, a single underlying concept (here, membership in one percolation cluster) getting represented by several dictionary atoms instead of one [Bricken et al., 2023, Cunningham et al., 2023], would then test whether that same loop content propagates three steps downstream, into a trained model’s own learned features.

A preliminary campaign along these lines has been run but is not reported here in detail. Two independent passes at different scales agreed on the qualitative direction of the effect (both cycle inconsistency and feature-splitting score fell monotonically with q) but disagreed on absolute magnitude by more than sampling noise should allow, most plausibly because cluster sampling, embedding dimension, and training configuration were not held fixed between the two passes. Reconciling this under one fixed, controlled configuration, rather than reporting whichever numbers a particular run happened to produce, is necessary before this pipeline’s output should be treated as a result rather than a promising direction, and is left to future work.

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